

Statistical Machine Learning

Assignment 4

AdaBoost · Gradient Boosting · Rosenblatt Perceptron

April 2026
IIIT Delhi

Question 1: AdaBoost with Decision Stumps [2 Marks]

1.1 Setup and Preprocessing

The MNIST dataset was loaded and filtered to retain only classes 4 and 9, labeled as -1 and $+1$ respectively. From the training set, 1000 samples per class were held out to form the validation set, ensuring no data leakage. PCA was applied on the remaining training samples to reduce the feature dimensionality from 784 to 5 principal components. Crucially, the PCA transformation matrix was computed solely on the training set and then applied to the validation and test sets.

1.2 AdaBoost Algorithm

A decision stump ensemble was grown using the AdaBoost algorithm. For each stump, the best split was determined by evaluating candidate thresholds — specifically midpoints between consecutive sorted unique values for each of the 5 dimensions (randomly sampled to at most 1000 per dimension for efficiency). The stump that minimized the weighted misclassification error was selected at each iteration. The stump weight α_t and sample weight update followed the standard AdaBoost formulation:

$$\alpha_t = 0.5 \cdot \ln((1 - \epsilon_t) / \epsilon_t)$$

$$w_i \leftarrow w_i \cdot \exp(-\alpha_t \cdot y_i \cdot h_t(x_i)) \text{ (then normalized)}$$

A total of 300 stumps were grown sequentially, with the ensemble prediction at iteration T being the weighted sign vote: $H(x) = \text{sign}(\sum \alpha_t \cdot h_t(x))$.

1.3 Results

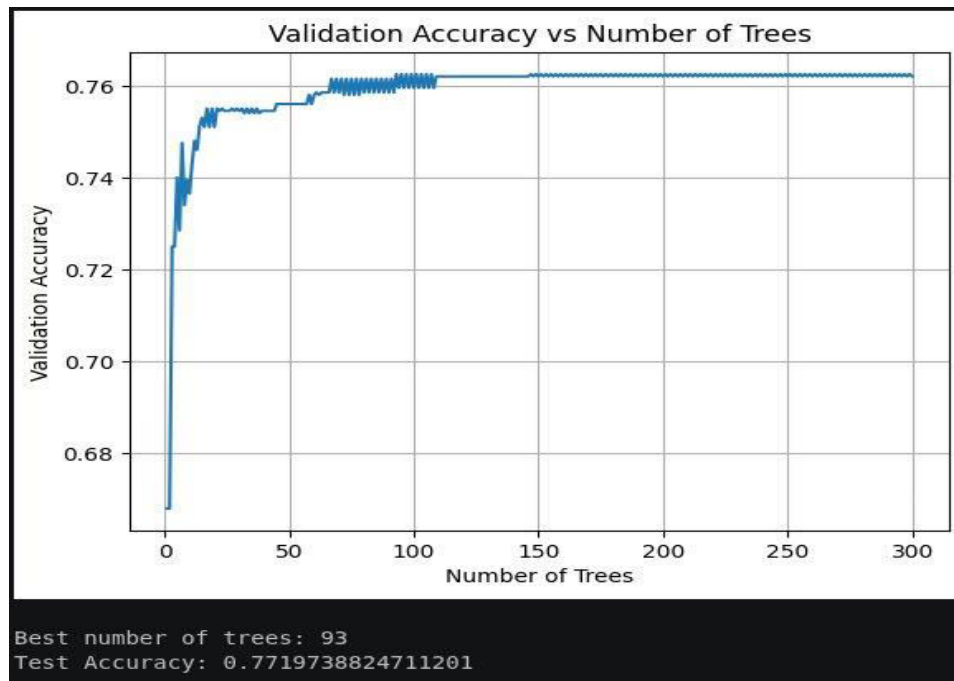


Figure 1.1: Validation Accuracy vs. Number of Trees (AdaBoost)

The validation accuracy improves sharply in the first ~30 iterations, rising from approximately 0.66 to 0.75, and then continues to increase more gradually until plateauing around 0.761 after 93 trees. Beyond this point, the ensemble accuracy remains stable, demonstrating that AdaBoost converges robustly without significant overfitting.

Metric	Value
Best Number of Trees	93
Test Accuracy	77.20%

1.4 Analysis

The initial steep rise in validation accuracy reflects the power of AdaBoost to quickly exploit the most discriminative splits in the PCA-reduced space. The plateau behavior indicates that 93 stumps already capture the bulk of learnable signal in 5 dimensions. The test accuracy of ~77.2% is consistent with the validation accuracy, confirming that the model generalizes well and the validation-based early stopping criterion was effective.

The modest absolute accuracy (~77%) is expected given that only 5 principal components are retained — a significant dimensionality reduction from the original 784-dimensional space.

Question 2: Gradient Boosting with Decision Stumps [2 Marks]

2.1 Setup and Preprocessing

The same dataset and PCA preprocessing as Question 1 were used (MNIST classes 4 and 9, reduced to 5 dimensions). The task is formulated as regression-based gradient boosting for binary classification, where labels are -1 and $+1$.

2.2 Gradient Boosting Algorithm

At each iteration t , a decision stump $h_t(x)$ is fit to minimize the Sum of Squared Residuals (SSR) on the current residuals. For absolute loss, the negative gradient is the sign of the residual. The ensemble prediction is updated as:

$$F_t(x) = F_{t-1}(x) + \eta \cdot h_t(x)$$

where η is the learning rate. The residuals (labels for the next stump) are updated after each iteration as $y - F_t(x)$. This continues for 300 iterations.

2.3 Effect of Learning Rate

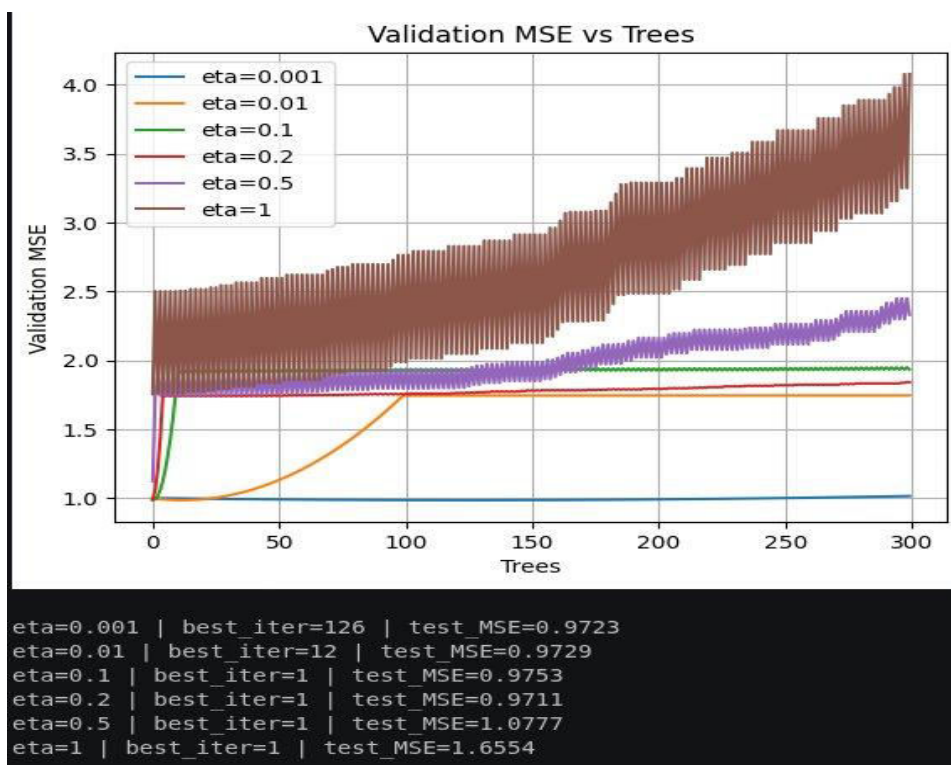


Figure 2.1: Validation MSE vs. Number of Trees for $\eta \in \{0.001, 0.01, 0.1, 0.2, 0.5, 1\}$

The results across different learning rates reveal clear trends:

- $\eta = 0.001$ (blue): Extremely slow convergence. The validation MSE decreases very gradually, reaching its best at iteration 126 (test MSE ≈ 0.9723). Despite 300 trees, the model is still improving and has not plateaued, indicating the learning rate is too small for this budget of trees.
- $\eta = 0.01$ (orange): More aggressive convergence, best at iteration 12 (test MSE ≈ 0.9729). The MSE rises sharply after the best iteration, indicating overfitting begins quickly at this learning rate.
- $\eta = 0.1$ (green): Best performance at iteration 1 (test MSE ≈ 0.9753). Very rapid initial fit but diverges afterward, staying around MSE ≈ 1.9 .
- $\eta = 0.2$ (red): Similar behavior to $\eta = 0.1$ — immediate best at iteration 1 with test MSE ≈ 0.9711 (lowest among all), then diverges.
- $\eta = 0.5$ (purple): High variance, oscillating MSE throughout 300 trees, diverging to ≈ 2.4 by the end.
- $\eta = 1.0$ (brown): Most aggressive — rapidly diverges with high oscillation, reaching MSE ≈ 4.1 by iteration 300.

η	Best Iteration	Test MSE	Behavior
0.001	126	0.9723	Slow, still converging
0.01	12	0.9729	Converges, then overfits
0.1	1	0.9753	Fast fit, then diverges
0.2	1	0.9711	Lowest test MSE (best)
0.5	1	1.0777	High variance / oscillates
1.0	1	1.6554	Diverges rapidly

2.4 Analysis

The key observations from the gradient boosting experiments are:

- Small learning rates ($\eta = 0.001$) require many trees to converge but are stable and consistent. They benefit from a larger tree budget.
- Moderate rates ($\eta = 0.01$) strike a balance — achieving good test MSE with far fewer trees (12), though validation MSE eventually rises due to overfitting.

- Large rates ($\eta \geq 0.1$) lead to immediate overfitting or divergence, with the exception that $\eta = 0.2$ achieves the best single-iteration test MSE of 0.9711. However, this reflects the ensemble collapsing to a single useful tree rather than genuinely good boosting.
- The general pattern confirms the well-known bias-variance trade-off in gradient boosting: smaller η requires more trees but typically generalizes better; larger η risks instability and divergence.

Question 3: Rosenblatt Perceptron [2 Marks]

3.1 Dataset Description

Two 2D Gaussian datasets were generated for binary classification:

- Dataset A: Class -1 sampled from $N([-3, -3]^T, I)$, Class $+1$ from $N([3, 3]^T, I)$. 200 samples per class, 70/30 train-test split.
- Dataset B: Same class means as Dataset A but with covariance $3I$ (higher spread). 200 samples per class, 70/30 train-test split.

All weights and biases were initialized to zero. The Rosenblatt perceptron was trained with learning rate $\eta = 0.01$ for a maximum of 300 epochs, with early stopping triggered if a full epoch passes with zero misclassifications.

3.2 Dataset A Results

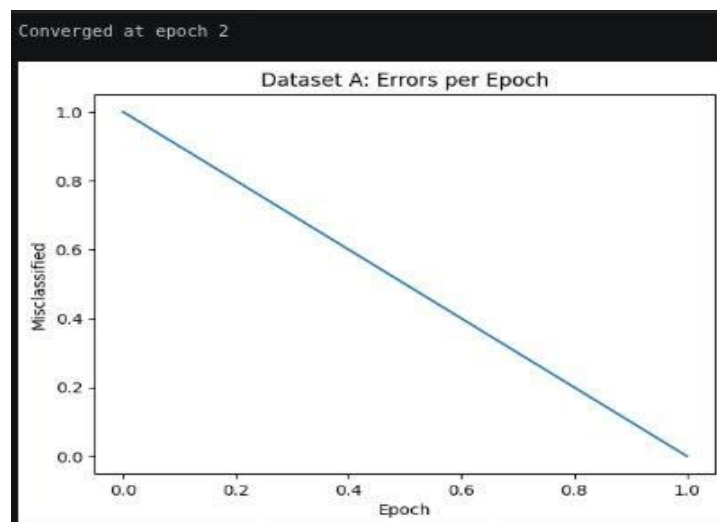


Figure 3.1: Dataset A — Misclassifications per Epoch

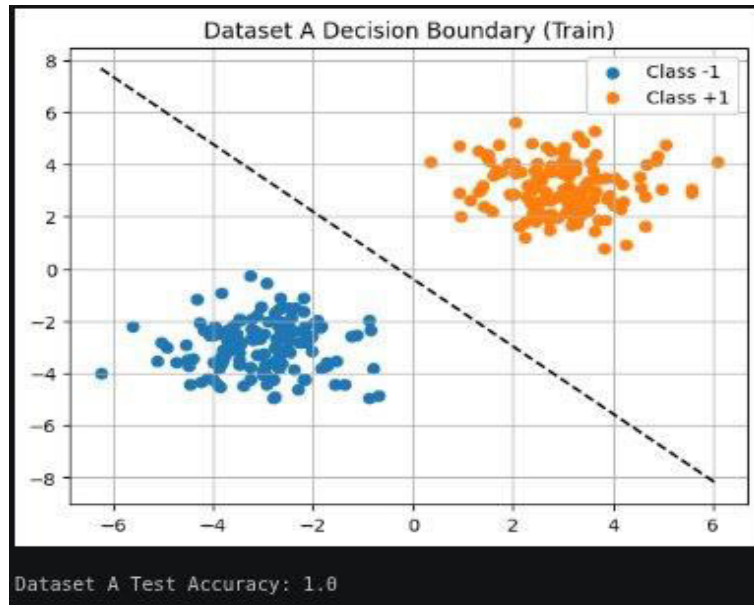


Figure 3.2: Dataset A — Decision Boundary (Train) | Test Accuracy: 1.0

Dataset A converged at epoch 2. The misclassifications decrease linearly from epoch 0 to epoch 1, reaching zero by epoch 2, triggering early stopping. The decision boundary learned is clean and well-separating, correctly classifying all test samples. The perfect test accuracy of 1.0 reflects the linearly separable nature of Dataset A — the two Gaussian clusters with unit covariance are well-separated with a large margin, making convergence trivial for the perceptron.

3.3 Dataset B Results

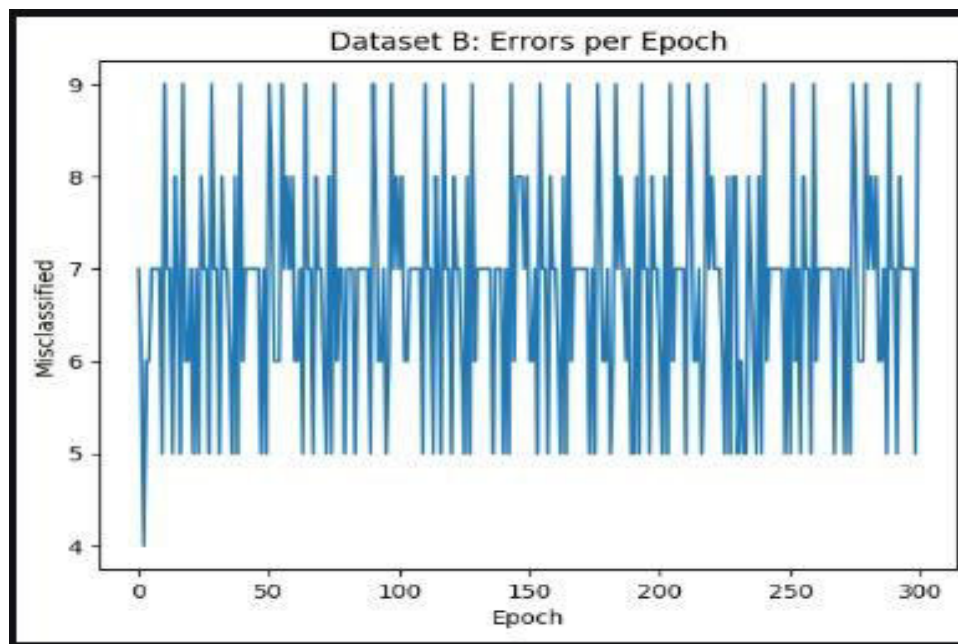


Figure 3.3: Dataset B — Misclassifications per Epoch

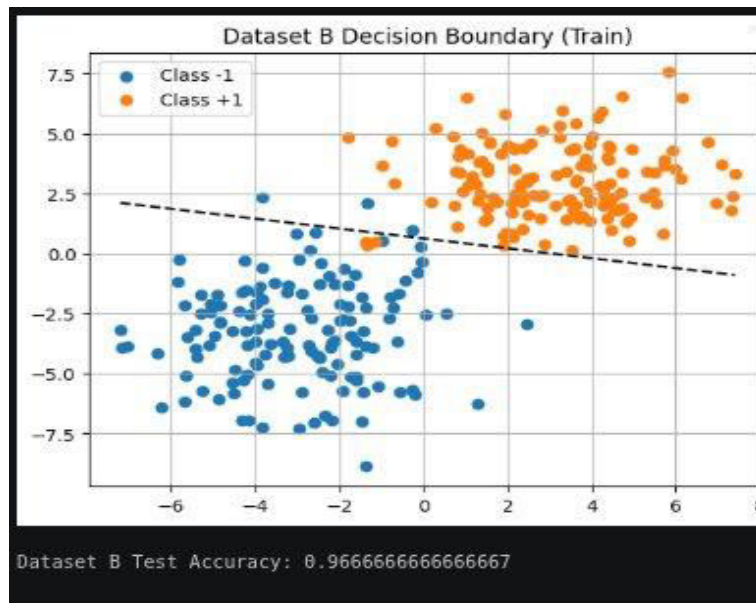


Figure 3.4: Dataset B — Decision Boundary (Train) | Test Accuracy: 0.967

Dataset B did not converge within the 300-epoch limit. The misclassifications oscillate erratically between approximately 4 and 9 throughout training, with no clear downward trend. Despite this, the perceptron achieves a test accuracy of 96.7%, and the plotted decision boundary provides a reasonable separation of the two classes.

3.4 Comparative Analysis

The contrasting behavior between Dataset A and Dataset B can be attributed to the difference in class overlap due to covariance:

- Dataset A (covariance I): The clusters are tight and well-separated. The data is linearly separable, so the Perceptron Convergence Theorem guarantees convergence in finite steps. Convergence is achieved in just 2 epochs.
- Dataset B (covariance 3I): The larger spread causes overlap between the two classes near the decision boundary. Since the classes are no longer cleanly linearly separable, the perceptron update rule oscillates — whenever it corrects one misclassified sample, it may cause another to be misclassified. This matches the theoretical result that the perceptron algorithm does not converge on non-linearly-separable data.
- The oscillating misclassification count in Dataset B is characteristic of the perceptron cycling behavior on overlapping distributions. The final boundary still generalizes reasonably well (96.7% test accuracy), but the training process itself never stabilizes.
- In practice, for Dataset B, a soft-margin classifier (e.g., SVM with slack variables) or a probabilistic model (e.g., logistic regression) would be more appropriate, as they are designed to handle non-separable data gracefully.

Property	Dataset A	Dataset B
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Covariance	Identity (I)	3I (high spread)
Linear Separability	Yes	No (overlapping)
Convergence	Epoch 2	Did not converge (300 epochs)
Misclassification Trend	Monotone decrease	Oscillating (4–9)
Test Accuracy	100%	96.7%

Summary

This assignment explored three fundamental ensemble and classification algorithms. AdaBoost (Q1) demonstrated rapid accuracy gains via weighted stumps, reaching peak validation performance at 93 trees with 77.2% test accuracy on MNIST classes 4 vs 9 in 5 PCA dimensions. Gradient Boosting (Q2) revealed a clear learning rate trade-off: small η (0.001–0.01) provides stable convergence over more iterations, while large η (≥ 0.5) causes divergence; the best single test MSE (0.9711) was achieved at $\eta = 0.2$ with just 1 tree, though this is not a practically meaningful result. The Rosenblatt Perceptron (Q3) confirmed the Perceptron Convergence Theorem in practice — converging in 2 epochs on the linearly separable Dataset A but failing to converge on the overlapping Dataset B, while still achieving 96.7% test accuracy through an approximately correct boundary.